Data Analysis Project

NBA Player Salaries

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STA 141A 2024 Summer Session 2

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Project Report

a.) Names And UCD Email Addresses:

- Winson Yu: wiyu@ucdavis.edu
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b.) Group Members' Contributions:

- Winson Yu:
  - Data Manipulation / Tidying
  - Programming Data Visualizations
  - Data Analysis
- Roy Ho:
  - Data communication

Introduction And Research Questions:

In the NBA, a player's salary can depend heavily on a wide range of factors, such as player performance, marketability, popularity, general management, and much more. For our project, we decided to focus on how a player's performance influences their respective salaries; specifically, the performance of both direct metrics such as points per game (PPG), assists per game (APG), and rebounds per game (RPG), as well as advanced metrics such as player efficiency (PER), effective field goal percentage (eFG%), and win shares (WS). We chose the direct metrics because those are the three most common statistics in basketball, and the advanced metrics were chosen because those three are less known and much less visible compared to the direct metrics. Based on our data set, we have posed these two questions:

- Do more tangible metrics of measuring player performance such as points per game, assists per game, and rebounds per game determine salary more accurately, or is it the advanced metrics such as player efficiency, effective field goal percentage, and win shares that influence salary the most?
- To what extent does the position of the player influence their salary?

Data Description:

The data we used was from Kaggle, which at first gave us two csv files: One for NBA player salaries spanning from 1985 to 2022, and another file with NBA player descriptions and performance details such as height, position, draft number, etc. We combined these two data sets into one file to make the data analysis easier. Initially, the salaries were by year, but we transformed the salaries across years into average salary over a career by taking the mean over the years active for corresponding players because the performance data was measured by game (e.g. points per game).

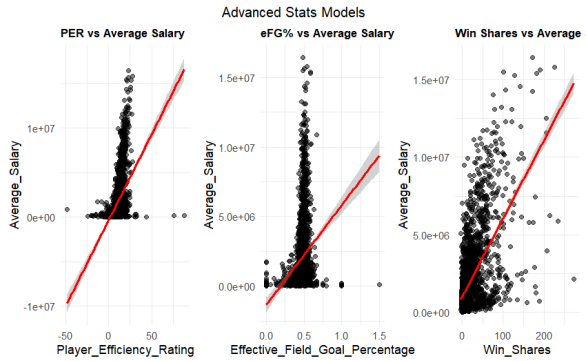
Data Visualization / Analysis / Methodology

For our data visualizations, we created several plots which visualize the relationship between the average salary that a particular NBA player receives over the course of career and several of our predictor variables. Below is a grid view of a few of the plots that show the general trend of the data for our basic linear regression models before transformations.

To answer our first question, we will attempt to fit linear regression models to each of our predictor variables and investigate the R^2 value and p-value to determine the precision of the fit. We addressed lack of linearity by attempting to fit our models to transformations such as log, square root and polynomial. To test the accuracy of the fit, we examine the R^2 value and the

p-values to explore the statistical significance and the correlation between the predictors and response.

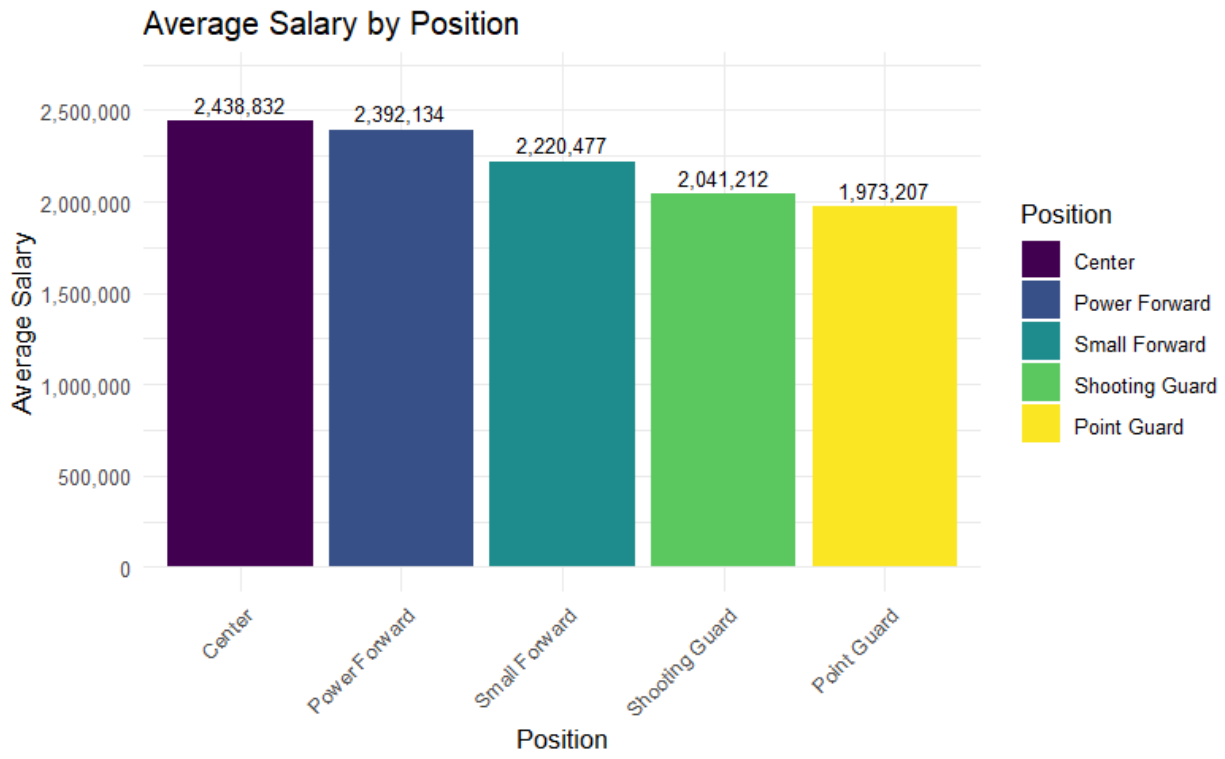
To answer our second question, we create a bar plot to visualize if there exists a major discrepancy between the salaries based on player position.

	
These plots illustrate the relationship between points, rebounds, and assists, demonstrating a positive correlation with salary.	These plots illustrate that player efficiency and effective field goal percentage are less efficient predictors of salary compared to the previous more direct methods of predicting salary, however, win share seems to provide a good fit with salary.

In conclusion, generally speaking, direct metrics, as shown with our 6 plots, tend to have a stronger influence on salary compared to advanced metrics.

Our second question was regarding whether the position of an NBA player has any influence on salaries. The bar plot below shows average salaries of NBA players by position. We see that on average, centers make the most money out of all of the positions at 2.4 million

dollars, followed by power forwards, small forwards, shooting guards, and point guards coming in last with an average salary of around 1.9 million dollars.



Conclusion and Discussion of Results

R^2 table

transform	direct	points	rebounds	assists	advanced	efficiency	fg%	winshare
none	0.4458	0.4106	0.312	0.1578	0.3882	0.1779	0.06181	0.3705
log	0.396	0.3478	0.3005	0.1303	0.2978	0.1644	0.09678	0.2606
$\sqrt{\ }$	0.4661	0.4192	0.3406	0.1599	0.3736	0.1877	0.08549	0.3466
poly		0.4255	0.3611	0.1794		0.193	0.09679	0.4142

The conclusion of our first question is that we reached for our first question involves referencing the R^2 values as shown in the table above. We observe that the regression fit does not improve under response variable transformations. This implies that the model itself may be

inefficient for predicting the salaries of NBA players overall, and there are several other factors as discussed in the introduction that need to be addressed to better predict the salary of an NBA player. However, we can also see that of our analyzed predictors, the points, rebounds, and win share seem to be the most significant of the variables that we explored to predict NBA player salary by their greater R^2 value.

Additionally, we explored the p-values which can be examined in the separate R markdown file. We observe that all of the predictors are statistically significant, that is to say, their p-value is extremely small which suggests that there is evidence to reject the null hypothesis of $\beta_x = 0$ for our predictor variables x, implying that there is a relationship between our predictors and response.

To conclude, our models suggested a relationship between our predictors and response, but there was still a significant portion of the variance was not accounted for when analyzing the R^2 value. This suggests that there may be many other factors that influence the salary of an NBA player that our model overlooks, but out of our options, we found that the best predictors were points, rebounds, and win share, implying that direct and advanced metrics are just a piece of the puzzle towards fully predicting the salary of an NBA player.

Our second key question of a player's position's influence on salary is also seen clearly with our bar plot. We see that on average, centers make the most out of any position in the NBA, averaging around 2.4 million dollars per year, with point guards averaging the least amount of money made, coming in at around 1.9 million dollars per year.

Completed Data Analysis:

Throughout our dataset, we had to perform logarithmic, square root, and polynomial transformations to the average salary so that non-linear relationships were properly accounted for

between our variables. In doing so, we organized the following R squared values by direct metrics and advanced metrics. We see clearly how the highest R squared values rested in the direct metrics, whereas the lowest R squared values were in those of the advanced metrics. This shows that generally, at the highest R squared value, around 44% of the variation of the response variable, salary, is explained by the predictors of points per game, assists per game, and rebounds per game. When it comes to the highest R squared value from the advanced metrics it showed that win shares held that value, which was expected based on what we saw from the win shares graph. Additionally, with our VIF table, we see that all of the values for both the direct metrics and advanced metrics were below 5, confirming that there was no significant multicollinearity between the variables, meaning that no two predictors were highly correlated among each other. This further supports our data and shows that our regression coefficients are accurate.

Detailed Assessment of Key Questions:

The two key questions proposed in our project were whether direct metrics or advanced metrics influenced salary more, and whether the position of a player influenced their salary as well. Our results found that although advanced metrics do have an influence on salary, as seen with the three designated graphs, ultimately it is the direct metrics that have the most influence on salary. When looking at the R squared values of the direct and advanced metrics, the highest R squared values were in the table of the direct metrics, which showed that there was a greater variation of the response variable based on the direct metric predictors.

R Appendix:

```
library(car)
```

```r
library(dplyr)
library(tidyverse)
library(ggplot2)
library(gridExtra)
library(viridis)

# Load Player data set
player_stats <- read.csv("../dataset/players.csv")

# Load Salaries data set, take mean salary over entire career
salaries <- read.csv("../dataset/salaries_1985to2018.csv")

calculate_average_salaries <- function(data) {
  # Ensure the data is in a data frame
  data <- as.data.frame(data)
  # Calculate average salary for each player
  result <- aggregate(salary ~ player_id, data, mean)
  # Rename the salary column to average_salary
  names(result)[names(result) == "salary"] <- "average_salary"
  # Arrange by player_id
  result <- result[order(result$player_id), ]
  return(result)
}

calculate_sum_salaries <- function(data) {
  # ensure data is in a data frame
  data <- as.data.frame(data)
  # Calculate average salary for each player
  result <- aggregate(salary ~ player_id, data, sum)
  # Rename the salary column to average_salary
  names(result)[names(result) == "salary"] <- "total_salary"
  # Arrange by player_id
  result <- result[order(result$player_id), ]
  return(result)
}

average_salaries <- calculate_average_salaries(salaries)
total_salaries <- calculate_sum_salaries(salaries)
```

```
#Combine both data sets by the index
combined_data <- player_stats %>%
  left_join(average_salaries, by = c("X_id" = "player_id"))

combined_data2 <- combined_data %>%
  left_join(total_salaries, by = c("X_id" = "player_id"))

# Select the necessary data columns, change column names
select_data <- combined_data2 %>%
  mutate(
    Average_Salary = average_salary,
    Total_Salary = total_salary,
    Points = as.numeric(career_PTS),
    Assists = as.numeric(career_AST),
    Rebounds = as.numeric(career_TRB),
    Player_Efficiency_Rating = as.numeric(career_PER),
    Effective_Field_Goal_Percentage = as.numeric(gsub("%", "",
career_eFG.)) / 100,
    Win_Shares = as.numeric(career_WS),
    Player_Position = strsplit(as.character(position), " and "),
  ) %>%
  select(name, Player_Position, Average_Salary, Total_Salary, Points,
Assists, Rebounds, Player_Efficiency_Rating,
Effective_Field_Goal_Percentage, Win_Shares)
colnames(select_data)[1] <- c("Name")

# Remove rows containing NA, sort data according to name
data_nona <- select_data %>%
  na.omit()
sorted_data <- data_nona %>% arrange(Name)
data <- sorted_data

#Fit linear regression in terms of points, rebounds, assists
direct_model <- lm(formula = Average_Salary ~ Points + Rebounds +
Assists, data = data)
summary(direct_model)
vif(direct_model)

data$log_Avg_Salary <- log(data$Average_Salary)
```

```
direct_model_log <- lm(log_Avg_Salary ~ Points + Rebounds + Assists,
data = data)
summary(direct_model_log)

data$sqrt_Avg_Salary <- sqrt(data$Average_Salary)
direct_model_sqrt <- lm(sqrt_Avg_Salary ~ Points + Rebounds +
Assists, data = data)
summary(direct_model_sqrt)

# Points
points_model <- lm(Average_Salary ~ Points, data = data)
summary(points_model)

points_model_log <- lm(log_Avg_Salary ~ Points, data = data)
summary(points_model_log)

points_model_sqrt <- lm(sqrt_Avg_Salary ~ Points, data = data)
summary(points_model_sqrt)

points_model_poly <- lm(sqrt_Avg_Salary ~ Points + I(Points^2), data
= data)
summary(points_model_poly)

# Rebounds
rebounds_model <- lm(Average_Salary ~ Rebounds, data = data)
summary(rebounds_model)

rebounds_model_log <- lm(log_Avg_Salary ~ Rebounds, data = data)
summary(rebounds_model_log)

rebounds_model_sqrt <- lm(sqrt_Avg_Salary ~ Rebounds, data = data)
summary(rebounds_model_sqrt)

rebounds_model_poly <- lm(sqrt_Avg_Salary ~ Rebounds + I(Rebounds^2),
data = data)
summary(rebounds_model_poly)

# Assists
assists_model <- lm(Average_Salary ~ Assists, data = data)
```

```
summary(assists_model)

assists_model_log <- lm(log_Avg_Salary ~ Assists, data = data)
summary(assists_model_log)

assists_model_sqrt <- lm(sqrt_Avg_Salary ~ Assists, data = data)
summary(assists_model_sqrt)

assists_model_poly <- lm(sqrt_Avg_Salary ~ Assists + I(Assists^2) +
I(Assists^3), data = data)
summary(assists_model_poly)

# Fit linear regression in terms of Efficiency, Field Goal %,
Win_Shares
advanced_model <- lm(formula = Average_Salary ~
Player_Efficiency_Rating + Effective_Field_Goal_Percentage +
Win_Shares, data = data)
summary(advanced_model)
vif(advanced_model)

advanced_model_log <- lm(log_Avg_Salary ~ Player_Efficiency_Rating +
Effective_Field_Goal_Percentage + Win_Shares, data = data)
summary(advanced_model_log)

advanced_model_sqrt <- lm(sqrt_Avg_Salary ~ Player_Efficiency_Rating
+ Effective_Field_Goal_Percentage + Win_Shares, data = data)
summary(advanced_model_sqrt)

# Player Efficiency Rating
efficiency_model <- lm(Average_Salary ~ Player_Efficiency_Rating,
data = data)
summary(efficiency_model)

efficiency_model_log <- lm(log_Avg_Salary ~ Player_Efficiency_Rating,
data = data)
summary(efficiency_model_log)

efficiency_model_sqrt <- lm(sqrt_Avg_Salary ~
Player_Efficiency_Rating, data = data)
```

```
summary(efficiency_model_sqrt)

efficiency_model_poly <- lm(sqrt_Avg_Salary ~
Player_Efficiency_Rating + I(Player_Efficiency_Rating^2), data =
data)
summary(efficiency_model_poly)

# Effective Field Goal Percentage
field_goal_model <- lm(Average_Salary ~
Effective_Field_Goal_Percentage, data = data)
summary(field_goal_model)

field_goal_model_log <- lm(log_Avg_Salary ~
Effective_Field_Goal_Percentage, data = data)
summary(field_goal_model_log)

field_goal_model_sqrt <- lm(sqrt_Avg_Salary ~
Effective_Field_Goal_Percentage, data = data)
summary(field_goal_model_sqrt)

field_goal_model_poly <- lm(sqrt_Avg_Salary ~
Effective_Field_Goal_Percentage +
I(Effective_Field_Goal_Percentage^2), data = data)
summary(field_goal_model_poly)

# Win Shares
ws_model <- lm(Average_Salary ~ Win_Shares, data = data)
summary(ws_model)

ws_model_log <- lm(log_Avg_Salary ~ Win_Shares, data = data)
summary(ws_model_log)

ws_model_sqrt <- lm(sqrt_Avg_Salary ~ Win_Shares, data = data)
summary(ws_model_sqrt)

ws_model_poly <- lm(sqrt_Avg_Salary ~ Win_Shares + I(Win_Shares^2),
data = data)
summary(ws_model_poly)
```

```r
# Salary by Position
create_model_plot <- function(model, x_var, y_var, title) {
  ggplot(data, aes_string(x = x_var, y = y_var)) +
    geom_point(alpha = 0.5) +
    geom_smooth(method = "lm", formula = y ~ x, color = "red") +
    labs(title = title, x = x_var, y = y_var) +
    theme_minimal() +
    theme(plot.title = element_text(size = 10, face = "bold"))
}

plot_points <- create_model_plot(points_model, "Points",
"Average_Salary", "Points vs Average Salary")
plot_rebounds <- create_model_plot(rebounds_model, "Rebounds",
"Average_Salary", "Rebounds vs Average Salary")
plot_assists <- create_model_plot(assists_model, "Assists",
"Average_Salary", "Assists vs Average Salary")

plot_per <- create_model_plot(efficiency_model,
"Player_Efficiency_Rating", "Average_Salary", "PER vs Average
Salary")
plot_efg <- create_model_plot(field_goal_model,
"Effective_Field_Goal_Percentage", "Average_Salary", "eFG% vs Average
Salary")
plot_ws <- create_model_plot(ws_model, "Win_Shares",
"Average_Salary", "Win Shares vs Average Salary")

basic_stats_plots <- grid.arrange(plot_points, plot_rebounds,
plot_assists, ncol = 3, top = "Basic Stats Models")
advanced_stats_plots <- grid.arrange(plot_per, plot_efg, plot_ws,
ncol = 3, top = "Advanced Stats Models")

all_plots <- grid.arrange(basic_stats_plots, advanced_stats_plots,
direct_model_plots, advanced_model_plots, ncol = 1)

# Position
avg_salaries <- data %>%
  unnest(Player_Position) %>%
  group_by(Player_Position) %>%
  summarize(avg_salary = mean(Average_Salary, na.rm = TRUE))
```

```r
salary_by_position <- avg_salaries %>%
  arrange(desc(avg_salary)) %>%
  mutate(Player_Position = factor(Player_Position, levels =
Player_Position))

salary_by_position

ggplot(data = salary_by_position) +
  geom_col(mapping = aes(x = Player_Position, y = avg_salary, fill =
Player_Position)) +
  geom_text(aes(x = Player_Position, y = avg_salary,
                label = scales::comma(round(avg_salary, 0))),
            vjust = -0.5, hjust = 0.5, size = 3, angle = 0) +
  labs(title = "Average Salary by Position",
       x = "Position",
       y = "Average Salary",
       fill = "Position") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_y_continuous(labels = scales::comma,
                     breaks = seq(0, 2500000, by = 500000),
                     limits = c(0, max(salary_by_position$avg_salary)
* 1.1)) +
  scale_fill_viridis_d()
```